

Monkies

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1 Matrix exercises

Exercise 1. Consider the following 3×3 matrix:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 3 & -2 & 1 \\ 4 & 1 & 1 \end{pmatrix}.$$

- a) Calculate the determinant of A (you may use Python directly). Why does the matrix have an inverse?
- b) Using the Gauss-Jordan method, calculate the inverse matrix of A , showing each step (you may use Python directly for the row operations).

Here is a label for the proof.

a) The determinant of A is given by

$$\begin{aligned} \det(A) &= \begin{vmatrix} 1 & 2 & 3 \\ 3 & -2 & 1 \\ 4 & 1 & 1 \end{vmatrix} \\ &= 1 \begin{vmatrix} -2 & 1 \\ 1 & 1 \end{vmatrix} - 2 \begin{vmatrix} 3 & 1 \\ 4 & 1 \end{vmatrix} + 3 \begin{vmatrix} 3 & -2 \\ 4 & 1 \end{vmatrix} \\ &= -3 + 2 + 33 = 32 \end{aligned}$$

Therefore, A is invertible since $\det(A) \neq 0$.

b) Since A is invertible, it follows that

$$(A|I_3) \sim (I_3|A^{-1}).$$

Therefore, by applying row operations

$$\begin{aligned}
 \begin{pmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 3 & -2 & 1 & | & 0 & 1 & 0 \\ 4 & 1 & 1 & | & 0 & 0 & 1 \end{pmatrix} &\sim \begin{pmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 0 & -8 & -8 & | & -3 & 1 & 0 \\ 4 & 1 & 1 & | & 0 & 0 & 1 \end{pmatrix} & F_2 - 3F_1 \rightarrow F_2, \\
 &\sim \begin{pmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 0 & -8 & -8 & | & -3 & 1 & 0 \\ 0 & -7 & -11 & | & -4 & 0 & 1 \end{pmatrix} & F_3 - 4F_1 \rightarrow F_3, \\
 &\sim \begin{pmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 0 & 1 & 1 & | & \frac{3}{8} & -\frac{1}{8} & 0 \\ 0 & -7 & -11 & | & -4 & 0 & 1 \end{pmatrix} & -\frac{1}{8}F_2 \rightarrow F_2, \\
 &\sim \begin{pmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 0 & 1 & 1 & | & \frac{3}{8} & -\frac{1}{8} & 0 \\ 0 & 7 & 11 & | & 4 & 0 & -1 \end{pmatrix} & -F_3 \rightarrow F_3 \\
 &\sim \begin{pmatrix} 1 & 0 & 1 & | & \frac{1}{4} & \frac{1}{4} & 0 \\ 0 & 1 & 1 & | & \frac{3}{8} & -\frac{1}{8} & 0 \\ 0 & 7 & 11 & | & 4 & 0 & -1 \end{pmatrix} & F_1 - 2F_2 \rightarrow F_1 \\
 &\sim \begin{pmatrix} 1 & 0 & 1 & | & \frac{1}{4} & \frac{1}{4} & 0 \\ 0 & 1 & 1 & | & \frac{3}{8} & -\frac{1}{8} & 0 \\ 0 & 0 & 4 & | & \frac{11}{8} & \frac{7}{8} & -1 \end{pmatrix} & F_3 - 7F_2 \rightarrow F_3 \\
 &\sim \begin{pmatrix} 1 & 0 & 1 & | & \frac{1}{4} & \frac{1}{4} & 0 \\ 0 & 1 & 1 & | & \frac{3}{8} & -\frac{1}{8} & 0 \\ 0 & 0 & 1 & | & \frac{11}{32} & \frac{7}{32} & -\frac{1}{4} \end{pmatrix} & \frac{1}{4}F_3 \rightarrow F_3
 \end{aligned}$$

$$\begin{aligned}
& \sim \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & \frac{1}{4} & \frac{1}{4} & 0 \\ 0 & 1 & 0 & \frac{1}{32} & -\frac{11}{32} & \frac{1}{4} \\ 0 & 0 & 1 & \frac{11}{32} & \frac{7}{32} & -\frac{1}{4} \end{array} \right) & F_2 - F_3 \rightarrow F_2 \\
& \sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{3}{32} & \frac{1}{32} & \frac{1}{4} \\ 0 & 1 & 0 & \frac{1}{32} & -\frac{11}{32} & \frac{1}{4} \\ 0 & 0 & 1 & \frac{11}{32} & \frac{7}{32} & -\frac{1}{4} \end{array} \right) & F_1 - F_3 \rightarrow F_1.
\end{aligned}$$

Therefore, the inverse matrix of matrix A is given by

$$A^{-1} = \begin{pmatrix} -\frac{3}{32} & \frac{1}{32} & \frac{1}{4} \\ \frac{1}{32} & -\frac{11}{32} & \frac{1}{4} \\ \frac{11}{32} & \frac{7}{32} & -\frac{1}{4} \end{pmatrix}.$$

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